Lecture 10: Comparing two treatments

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Outline

01

INTRODUCTION

From a business question to one parameter, and two ways of collecting the data

02

INDEPENDENT RANDOM SAMPLES

Large samples, pooled t, the F-test for equal variances, and Welch's t

03

MATCHED PAIR COMPARISONS

Analysing differences within pairs, and when pairing pays off

04

COMPARING TWO PROPORTIONS

The difference of two proportions, and what randomisation adds to a conclusion

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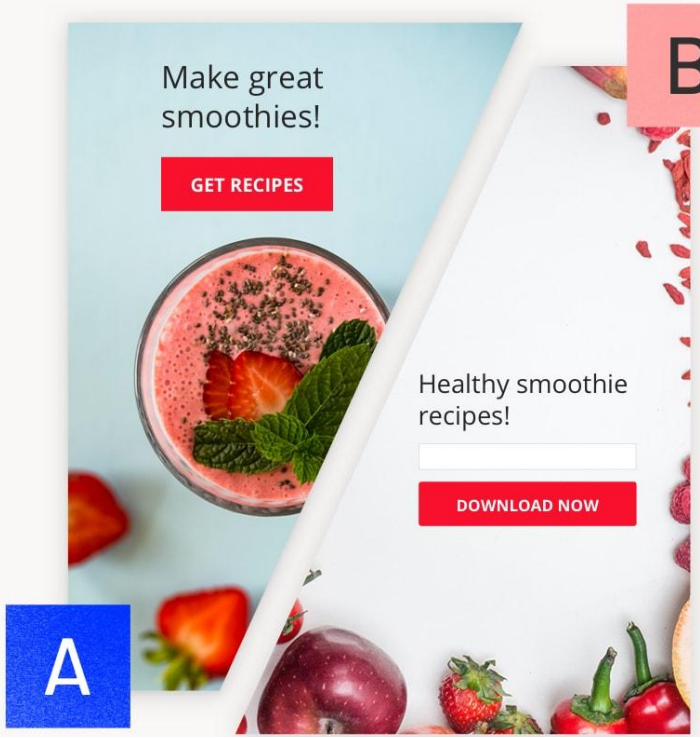
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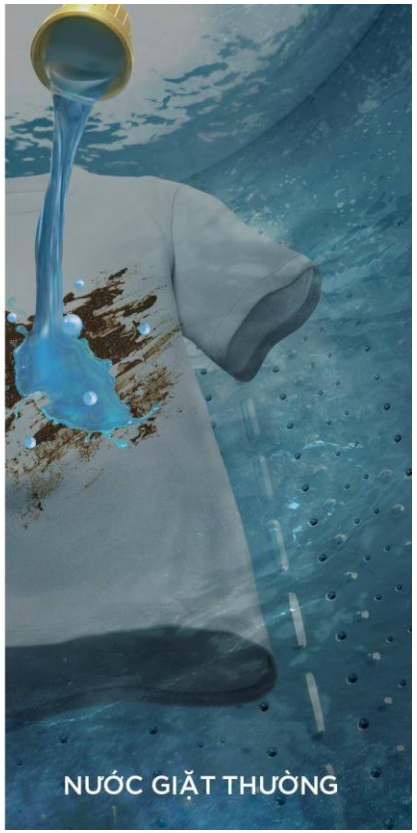

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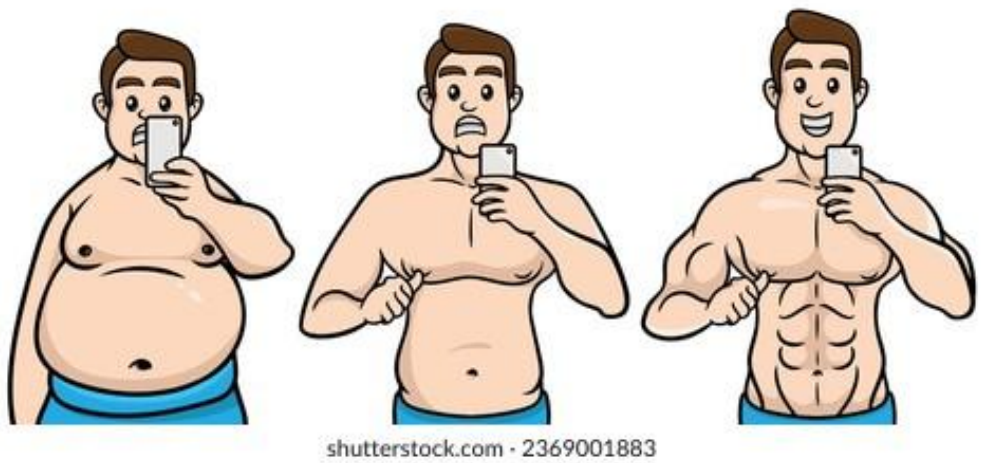
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INTRODUCTION

From a Business Question to a Parameter

A retail chain wants to know whether a new store layout raises weekly sales

"The new layout is better" becomes: mean weekly sales under the new layout exceed mean weekly sales under the old one.

The parameter is $\mu_1 - \mu_2$, and $H_0 : \mu_1 - \mu_2 = 0$.

A comparison of two populations always reduces to estimating one number — their difference. The sample estimate is $\bar{x}_1 - \bar{x}_2$. Everything that follows in this session is about one question: how uncertain is that estimate?

Two populations, one parameter: the difference.

INTRODUCTION

Two Ways to Collect the Data

The same question can be answered by two different designs

DESIGN A — INDEPENDENT SAMPLES

Eight stores switch, eight other stores do not

Two separate groups of stores are observed. No store appears in both groups, so the two samples are unrelated to each other.

- Called a two-sample design

DESIGN B — MATCHED PAIRS

The same eight stores, before and after

Each store is measured twice, once under each layout. Every observation under the new layout has a partner under the old one.

- Called a matched-pair design

Both designs answer the same question. Will they give the same conclusion?

INTRODUCTION

The Vocabulary of a Comparison

01

Treatment

The condition being compared: the new layout, a training method, a version of a web page.

02

Experimental unit

The object that receives a treatment and is measured: a store, an employee, a website visitor.

03

Two-sample design

Each unit receives one treatment only; the two groups of units are independent.

04

Matched-pair design

Units come in pairs that are alike, or one unit is measured under both treatments.

INTRODUCTION

From One Population to Two

Remind the five steps of Lecture 9 are unchanged; only the estimate and its standard error change

	Lecture 9: one population	Lecture 10: two populations
Parameter	μ	$\mu_1 - \mu_2$
Estimate	$\bar{x}$	$\bar{x}_1 - \bar{x}_2$
Null Hypothesis H_0	$\mu = \mu_0$	$\mu_1 - \mu_2 = 0$
Standard error	$s/\sqrt{n}$	depends on the design
Statistic = (estimate − value under H_0) ÷ standard error		

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INDEPENDENT RANDOM SAMPLES

The Standard Error of a Difference

Independent samples of sizes n_1 and n_2

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Variances increased, even though the means are subtracted.

- Each sample mean carries its own sampling error, and the two errors are independent.
- When two independent random quantities are combined — added or subtracted — their variances add. The difference of two estimates is therefore always less precise than either estimate on its own; the errors never cancel.

Can you explain this using mathematics?

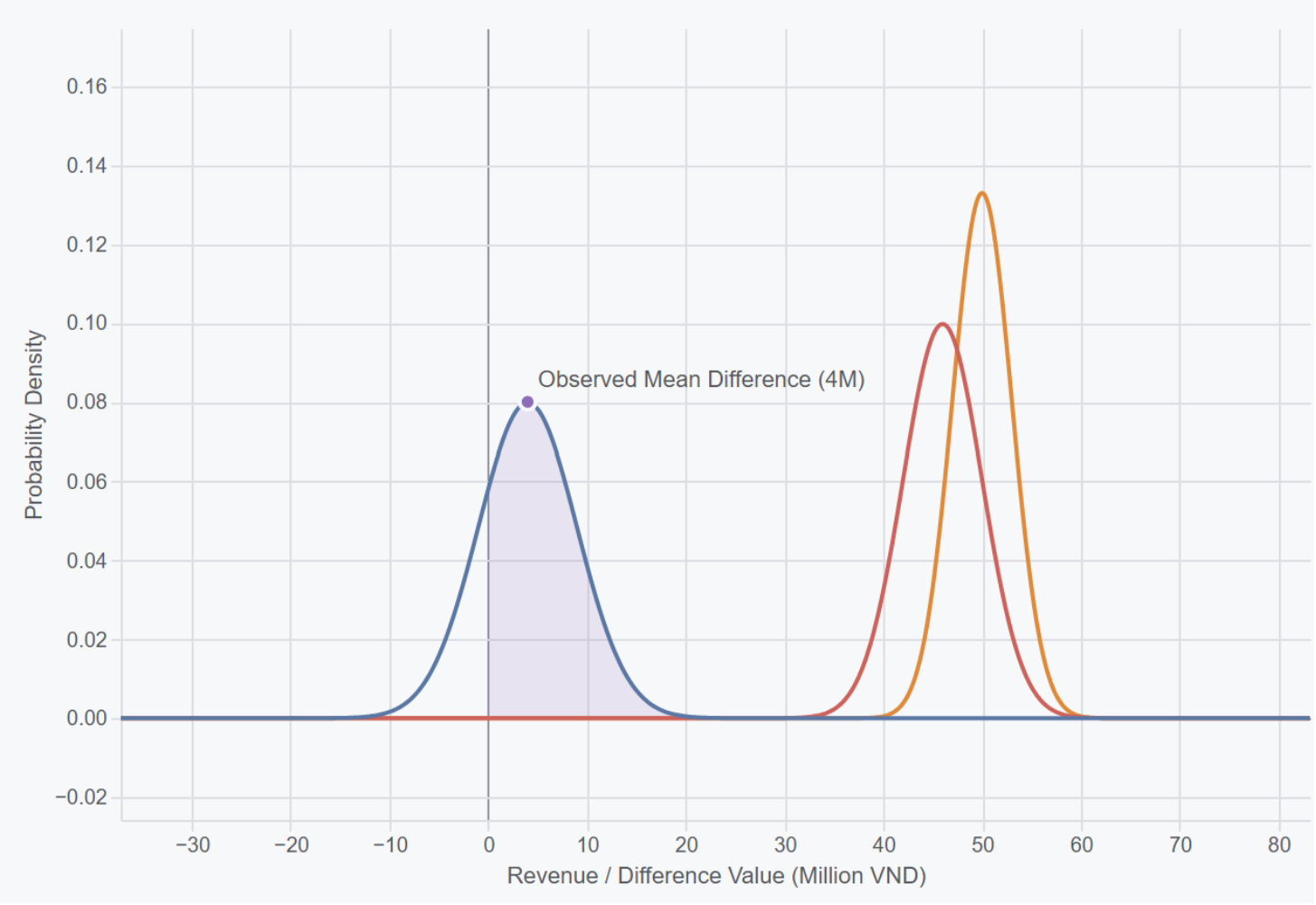
Hint: Var(X) = E(X^2) – E(X)^2

Example: Comparing 2 stores

Revenue Difference of 2 stores A & B

Store A: Centered at 50, narrower because its risk (standard deviation) is smaller (σ=3).
Store B: Centered at 46, slightly flatter and wider because it has higher volatility (σ=4).

Difference Curve (A - B) (Purple Curve): Centered at **4** (50−46=4). Notice that this curve is **much flatter and wider** than both individual curves



Three Cases, One Structure

Case	Statistic	Distribution
n_1 and n_2 both large	$Z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$	standard normal
small n, normal populations, $\sigma_1 = \sigma_2$	$T = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$	Student t-distribution with df = $n_1 + n_2 - 2$
small n, normal populations, $\sigma_1 \neq \sigma_2$	$\text{Welch's } t = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$	Student t-distribution with approximate df
The numerator is always $\bar{x}_1 - \bar{x}_2$. Only the denominator and the table change.		

Example: Order Value on the App and on the Web

An e-commerce platform compares the mean order value of app and web customers.

App: $n_1 = 100$, $\bar{x}_1 = 385$, $s_1 = 90$.

Web: $n_2 = 100$, $\bar{x}_2 = 352$, $s_2 = 120$

(All unit is in thousand VND).

Is there a difference in mean order value between the two channels? Test at $\alpha = 0.05$, then give a 95% confidence interval for $\mu_1 - \mu_2$.

Both samples are large, so the first case of the table applies.

The Order-Value Test, Step by Step

STEP 1

Hypotheses

$H_0 : \mu_1 - \mu_2 = 0$ versus $H_1 : \mu_1 - \mu_2 \neq 0$

STEP 2

Test statistic

$Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$, two-sided

STEP 3

Rejection region

$\alpha = 0.05$, so $R : |Z| \geq 1.96$

STEP 4

Compute

$SE = \sqrt{(8100/100 + 14400/100)} = \sqrt{225} = 15$. $Z = 33/15 = 2.20$

STEP 5

Conclude

$2.20 \geq 1.96$: H_0 is rejected. $p = 0.028$

95% interval: 33 ± 1.96 × 15 = [3.6 ; 62.4] thousand VND.

Statistical Significance vs Practical Significance

The test says the channels differ. The interval says by how much: somewhere between 3.6 and 62.4 thousand VND per order.

At the lower end the difference is too small to justify any change in marketing budget; at the upper end it would justify a major shift towards the app. Both are consistent with the data. The interval excludes 0 exactly when the two-sided test rejects at the same level — they are one computation — but only the interval reports the size of the difference.

Small Samples with Equal Variances: Pooling

If $\sigma_1 = \sigma_2 = \sigma$, both samples estimate the same σ, so they are combined:

$$s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$$

The pooled variance is a weighted average of the two sample variances, with weights equal to their degrees of freedom.

The statistic $T = \frac{\bar{x}_1-\bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$ has a t distribution with $n_1 + n_2 - 2$ degrees of freedom.

Pooling is valid only when the equal-variance assumption holds.

Example: Two Sales-Training Methods

Problem: A company wishes to evaluate and compare the effectiveness of two different sales training methods based on employee productivity scores.

Survey data from two independent groups of employees are recorded as follows:

Group 1 (trained using Method 1): 10 employees, mean productivity score of 78, sample standard deviation of 6.

Group 2 (trained using Method 2): 12 employees, mean productivity score of 72, sample standard deviation of 7.

Assume that productivity scores in both groups follow a normal distribution and that the two populations have equal variances

Requirements:

a, Hypothesis testing: At a significance level of $\alpha = 0.05$, test whether there is a statistically significant difference in mean productivity scores between the two training methods.

b, Interval estimation: Construct a 95% confidence interval for the difference in mean productivity scores between the two methods ($\mu_1 - \mu_2$) and state the practical significance of the result.

Example: Two Sales-Training Methods

Method 1: $n_1 = 10$, $\bar{x}_1 = 78$, $s_1 = 6$ · Method 2: $n_2 = 12$, $\bar{x}_2 = 72$, $s_2 = 7$ · productivity score, $\alpha = 0.05$

STEP 1

Hypotheses

$H_0 : \mu_1 - \mu_2 = 0$ versus $H_1 : \mu_1 - \mu_2 \neq 0$

STEP 2

Test statistic

pooled T, df = 10 + 12 − 2 = 20

STEP 3

Rejection region

R : $|T| \geq t_{0.025}(20) = 2.086$

STEP 4

Compute

$s_p^2 = (9 \cdot 36 + 11 \cdot 49)/20 = 43.15$. SE = $6.57 \cdot \sqrt{(1/10 + 1/12)} = 2.81$. T = 2.13

STEP 5

Conclude

$2.13 \geq 2.086$: H_0 is rejected, narrowly. p = 0.045

95% interval: 6 ± 2.086 × 2.81 = [0.13 ; 11.87] — close to 0, as the narrow rejection implies.

Checking the Assumption: The F-Test for Two Variances

$$H_0 : \sigma_1^2 = \sigma_2^2.$$

$$F = \frac{s^2_{\text{large}}}{s^2_{\text{small}}}$$

with (n − 1) degrees of freedom for each sample, numerator first. Reject when $F \geq F_{0.025}$.

The F distribution is the distribution of a ratio of two independent sample variances from normal populations. Like χ^2, it takes only positive values and is not symmetric.

The F-test is sensitive to non-normality. It is a check on the assumption, not a guarantee.

For the training example: $F = 49/36 = 1.36$, below $F_{0.025}(11, 9) = 3.91$, so equal variances are not rejected and pooling was justified.

Example: Daily Sales at Two Branches

Branch A: n = 20 days, $\bar{x}$ = 52, s = 5.

Branch B: n = 8 days, $\bar{x}$ = 45, s = 12 (million VND).

Is there a difference in mean daily sales? α = 0.05.

Check the assumption first: $F = 12^2/5^2 = 144/25 = 5.76$, which exceeds $F_{0.025}(7, 19) = 3.05$. Equal variances are rejected: branch B is far more volatile than branch A.

Conclusion: Pooling is not justified

Pooled t vs Welch's t on the Same Data

The same two samples, the same difference of 7, two different standard errors

	Pooled t (assumes $\sigma_1 = \sigma_2$)	Welch t (not assumes σ_1 equal to σ_2)
Standard error	s_p * √(1/20 + 1/8) = 3.16	√(25/20 + 144/8) = √19.25 = 4.39
T	7/3.16 = 2.22	7/4.39 = 1.60
df and critical value	26, 2.056	≈ 8, 2.306
p-value	0.036	0.149
Conclusion	reject H_o	do not reject H_o

A wrong assumption reversed the conclusion.

Why the Conclusion Flipped

MECHANISM

Pooling borrows precision from the calmer sample

The pooled variance weights each sample by its degrees of freedom. When the larger sample has the smaller variance, the pooled figure understates the noise contributed by the small, volatile sample.

- Worst case: small n together with large s

CONSEQUENCE

When in doubt, use Welch

If the variances are in fact equal, Welch and pooled give nearly the same answer, so little is lost. If they are not, pooled t declares differences that the data do not support.

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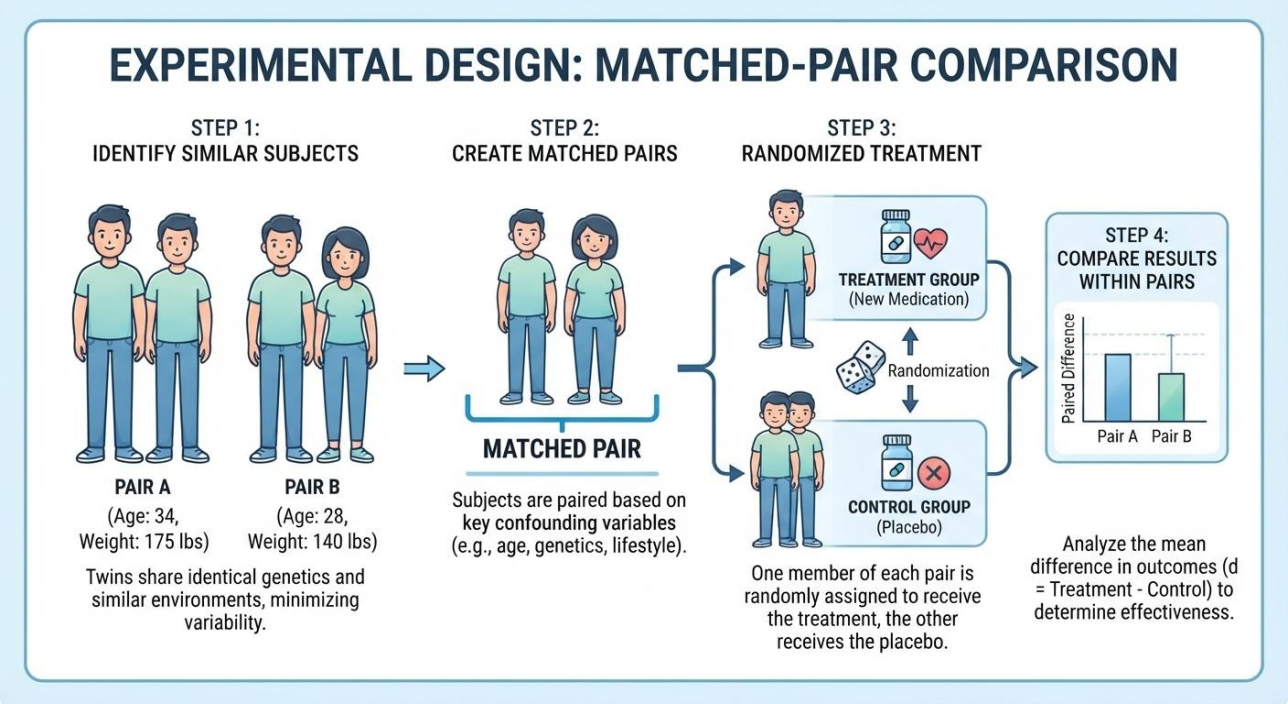
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COMPARING TWO PROPORTIONS

The difference of two proportions, and what randomisation adds to a conclusion

Matched Pairs Definition

- In comparing two treatments, it is desirable that the experimental units or subjects be **as alike as possible**, so that a difference in responses between the groups can be attributed to differences in treatments.
- But large variability can occur in the measurements.
- The procedure (sampling by matched pairs) consists of choosing units in pairs or blocks so that the units in each block are similar and those in different blocks are dissimilar.



How can we define 2 objects “matching”?

Matched Pairs: Analyze the Differences

For each pair compute $d_i = x_i(\text{after}) - x_i(\text{before})$.

Then $T = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$, with n − 1 degrees of freedom, and $H_0 : \mu_d = 0$.

This is the one-sample t-test of Lecture 9, applied to the column of differences. The two original columns are no longer analyzed separately; the unit of analysis is the change within each experimental unit. Whatever makes one unit differ from another is present in both of its measurements, and is removed by the subtraction.

No new formula — a new way of looking at the data.

MATCHED PAIR COMPARISONS

The Eight Stores

Weekly sales (million VND) before and after the new layout

Store	Before	After	d = After − Before
1 · 2	120 · 85	126 · 88	6 · 3
3 · 4	210 · 150	219 · 154	9 · 4
5 · 6	95 · 175	101 · 178	6 · 3
7 · 8	130 · 60	137 · 66	7 · 6
Mean	128.1	133.6	5.5

Sales range from 60 to 210 between stores. How large are the changes within stores?

MATCHED PAIR COMPARISONS

The Paired Test, Step by Step

STEP 1

Hypotheses

$H_0 : \mu_d = 0$ versus $H_1 : \mu_d \neq 0$

STEP 2

Test statistic

$T = \frac{\bar{d}}{s_d/\sqrt{n}}$, df = n − 1 = 7

STEP 3

Rejection region

R : $|T| \geq t_{0.025}(7) = 2.365$

STEP 4

Compute

$\bar{d}$ = 5.5, s_d = 2.07, SE = 2.07/√8 = 0.732.

T = 5.5/0.732 = 7.51

STEP 5

Conclude

7.51 ≥ 2.365: H_0 is rejected. p ≈ 0.0001

95% interval for μ_d : 5.5 ± 2.365 × 0.732 = [3.77 ; 7.23] million VND per week.

MATCHED PAIR COMPARISONS

The Same Sixteen Numbers, Analyzed Two Ways

The answer to the question left open in Part 1

Analysis	What goes into the standard error	T	p-value
Treated as two independent samples	differences between stores (s ≈ 50 in each column)	0.22	0.83
Treated as matched pairs	only the changes within stores (s_d = 2.07)	7.51	0.0001
The design decides what counts as noise			

MATCHED PAIR COMPARISONS

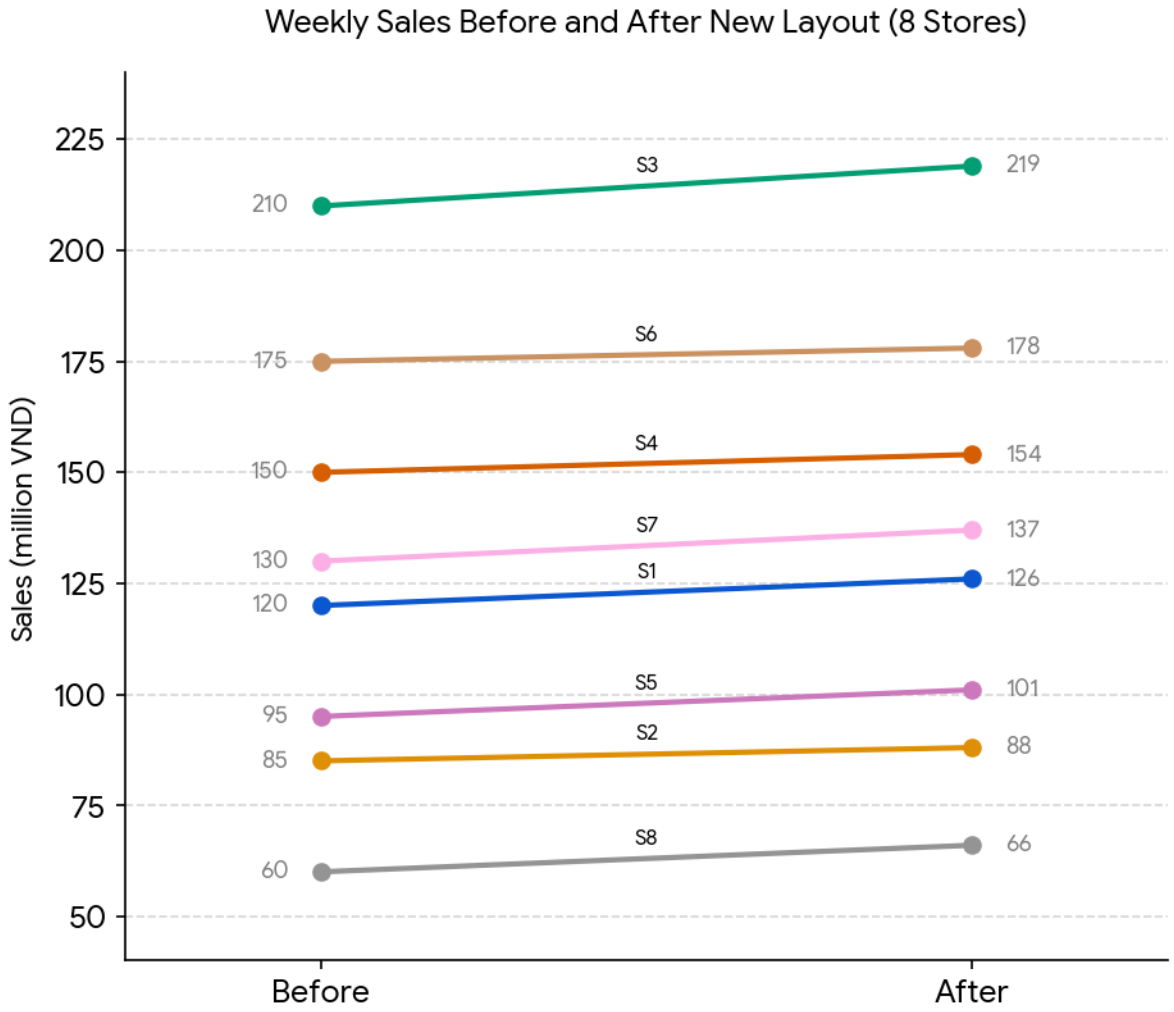
Seeing Why Pairing Wins

BEFORE AND AFTER

Eight lines, all rising

Each store is a line from its before value to its after value. The lines sit at very different heights, but every one of them slopes upward by a similar amount.

- Height of a line: differences between stores
- Slope of a line: the effect of the layout
- The paired test looks only at the slopes



$$\mathrm{Var}(X_{\text{after}} - X_{\text{before}}) = \mathrm{Var}(X_{\text{after}}) + \mathrm{Var}(X_{\text{before}}) - 2 \cdot \mathrm{Cov}(X_{\text{after}}, X_{\text{before}})$$

Independent or Paired: Choosing the Design

WHAT PAIRING GAINS

Variation between units leaves the error

Everything that makes one unit differ from another is removed by the subtraction. The standard error reflects only how consistently the treatment works within units.

- Worth it when units differ a lot compared with the effect

WHAT PAIRING COSTS

Degrees of freedom are halved

Eight pairs give 7 degrees of freedom instead of 14. If the pairs are not truly alike, pairing removes little variation and still pays this price in a larger critical value.

- Pairing is decided before the data are collected

MATCHED PAIR COMPARISONS

Matched Pairs in Business Data

Recognising a paired design is the first step of the analysis

01

Before and after

The same customers, stores or employees measured before and after a policy change.

02

One item, two channels

The same product's price or sales on two platforms, in the same week.

03

Two raters, one case

Two appraisers valuing the same set of properties, or two graders marking the same essays.

04

Matched units

Pairs of stores chosen to be alike in size and location, one of each pair receiving the treatment.

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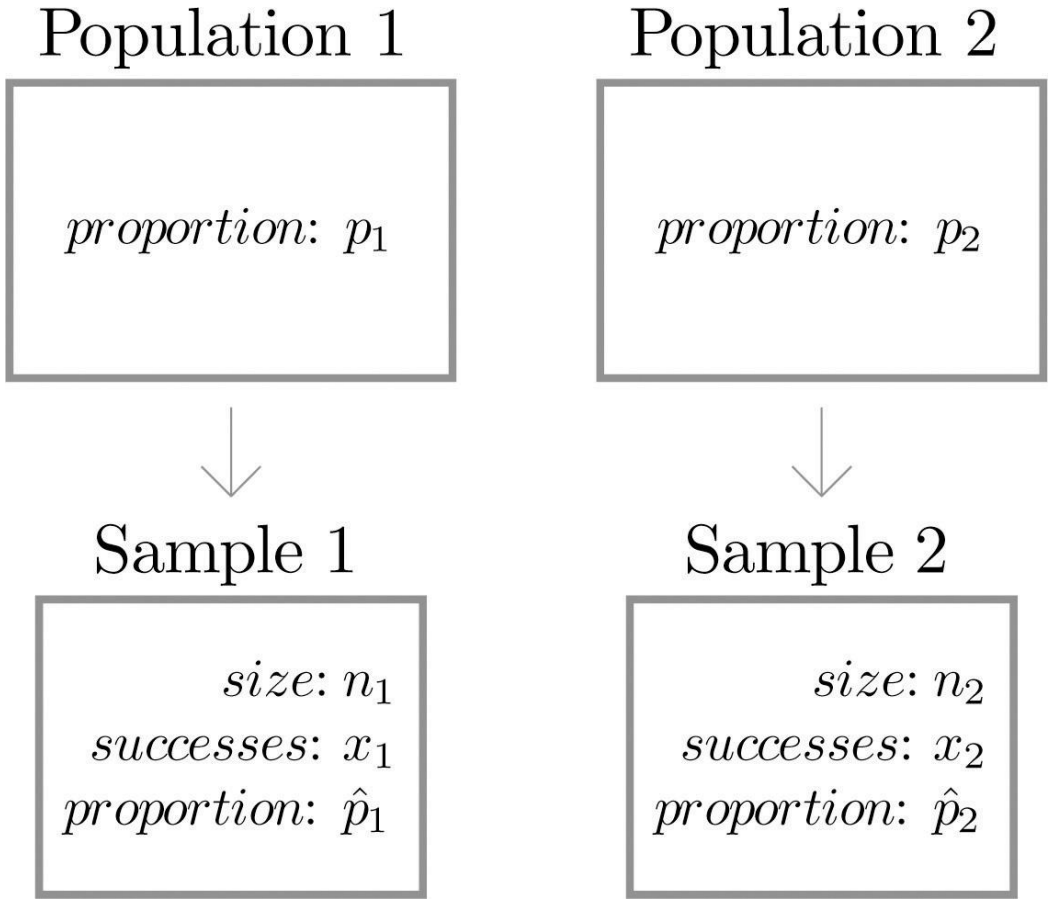
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Comparing Two Population Proportions

Comparing the jobless rates in two cities, the percentages of female employees in two categories of jobs, and infant mortality in two ethnic groups are just a few examples, etc. are often of our great interests

Let $p1$ denote the proportion of members possessing the characteristic in population 1, and $p2$ that in population 2. Our goals are to construct confidence intervals for $p1 - p2$ and test $H: p1 = p2$, the null hypothesis that the rates are the same for two populations.



Comparing Two Population Proportions

Parameter: $p_1 - p_2$. Estimate: $\hat{p}_1 - \hat{p}_2$, where $\hat{p} = X/n$. For large samples, $\hat{p}_1 - \hat{p}_2$ is approximately normal.

The structure is the one used for means: an estimated difference divided by its standard error.

The variance of a sample proportion is $p(1 - p)/n$, and for independent samples the two variances add.

Two Standard Errors for One Difference

The test and the interval use different standard errors, for a reason

Purpose	Standard error	Why
Test of $H_0 : p_1 = p_2$	$SE = \sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$ $\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$	the test is computed in the world of H_0, where there is one common p
Confidence interval for $p_1 - p_2$	$SE_{unpooled}$ $= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	an interval assumes nothing about H_0
Pooling here is required by H_0 — not an extra assumption, as it was for σ		

Example: An A/B Test of a Checkout Page

An online shop tests two versions of its checkout page. Visitors are assigned at random. Version A: 200 orders from 4,000 visits (5.0%). Version B: 248 orders from 4,000 visits (6.2%).

Does the conversion rate differ between the two versions? Test at $\alpha = 0.05$, then give a 95% confidence interval for $p_B - p_A$. Each group contains hundreds of both outcomes, so the normal approximation is safe.

COMPARING TWO PROPORTIONS

The A/B Test, Step by Step

STEP 1

Hypotheses

$H_0 : p_B - p_A = 0$ versus $H_1 : p_B - p_A \neq 0$

STEP 2

Test statistic

$$Z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

for pooled $\hat{p}$

STEP 3

Rejection region

$\alpha = 0.05$, so R : $|Z| \geq 1.96$

STEP 4

Compute

$\hat{p} = 448/8000 = 0.056$. SE = $\sqrt{(0.056 \cdot 0.944 \cdot 2/4000)} = 0.00514$. $Z = 2.33$

STEP 5

Conclude

$2.33 \geq 1.96$: H_0 is rejected. $p = 0.020$

How Large Is the Improvement?

95% interval for $p_B - p_A$: 1.2 ± 1.96 × 0.514 = [0.19 ; 2.21] percentage points, using the unpooled standard error.

The improvement is real at the 5% level, but its size is poorly determined: it may be as small as one-fifth of a point or as large as two points. Whether version B is worth rolling out depends on what the change costs — and a cost that is justified by 2 points may not be justified by 0.2.

The Role of Randomization

The same computation, applied to two situations

	A/B test of the checkout page	Dividend payers: HOSE versus HNX
How the two groups were formed	visitors assigned at random at A or B	firms chose their exchange, and met its listing rules
Is the test valid?	yes	yes
What may be concluded	version B causes a higher conversion rate	the exchanges differ in the proportion of dividend payers
What may not be concluded	—	"listing on HOSE makes firms pay dividends"
Causation is a property of how the data were collected, not of the formula		

The Role of Randomization

Random assignment makes the two groups alike, on average, in everything except the treatment. So that the difference between groups can be read as the effect of the treatment.

Where assignment cannot be randomized — the choice of exchange, a firm's industry, a customer's own decision to join a loyalty programme — a two-sample test still tells whether the groups differ. But it cannot say why.

Three Things to Take Away

1. SIGNAL OVER NOISE

Every comparison today is signal over noise: a difference over its standard error, or between-group over within-group variation.

2. THE DESIGN SETS THE ERROR

How the data were collected decides the standard error. The same sixteen numbers gave p = 0.83 unpaired and p = 0.0001 paired.

3. ONE QUESTION, THEN MANY

With k groups, ask once with ANOVA, then compare pairs at α/m. Only random assignment lets a difference be called an effect.